

Raj Thakur

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Professional Summary

AI Engineer with hands-on experience building LLM-powered applications, RAG pipelines, and computer vision systems. Proficient in Python, PyTorch, LangChain, FastAPI, and vector databases, with a track record of published IEEE research. Strong foundation in model evaluation, API design, and production deployment.

Education

Sage University
B.Tech in Computer Technology (Artificial Intelligence)

Indore, Madhya Pradesh
September 2022 – June 2026

Technical Skills

Languages: Python, SQL, JavaScript (basic)
ML/DL: PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Hugging Face Transformers
LLM/GenAI: LangChain, LlamaIndex, OpenAI API, RAG, Prompt Engineering, ChromaDB, FAISS, LoRA/PEFT
Backend/Data: FastAPI, REST APIs, PostgreSQL, MySQL, Pandas, NumPy, Docker
Tools: Git, GitHub, Postman, Linux, Jupyter, VS Code

Experience

AI Developer Intern June 2025 – July 2026
Parul Chemicals Vadodara, Gujarat

- Architected and built an internal AI-driven CRM using FastAPI, PostgreSQL, and LangChain, automating lead scoring and follow-up generation.
- Developed a sales/marketing assistant using fine-tuned NLP models and prompt-engineered LLM chains, reducing process time by 28%.
- Wrote modular, documented code with unit tests (pytest) and CI via GitHub Actions, improving maintainability and onboarding speed.

Projects

AI Resume Reviewer | *Python, LLM API, RAG, FastAPI* - GitHub May 2025

- Built a RAG-based resume analysis system using LangChain, Gemini, and FAISS, processing PDFs via PyPDF2 and generating structured feedback through Pydantic output parsers and function calling.
- Implemented recursive character chunking (512 tokens, 50-token overlap), text-embedding-3-small embeddings, and MMR-based retrieval to balance relevance and diversity in retrieved context.
- Designed evaluation across skills match, experience alignment, gap analysis, and ATS compatibility; reduced hallucination via retrieval-grounded prompting and JSON schema enforcement.

AI Invoice Automation & TallyPrime Integration | *PDF Processing, TallyPrime* June 2026

- Built an end-to-end invoice extraction pipeline using PDFPlumber for text extraction, regex for structured fields, and GPT-4o-mini as LLM fallback for unstructured invoices, with Pydantic validation before accounting entry.
- Achieved 98% field-level extraction accuracy across 80 test invoices (vendor, invoice no., dates, GST, amounts, line items); processed 20 invoices in 30–50 seconds end-to-end.
- Integrated with TallyPrime via its XML HTTP protocol and TDL for automated voucher creation; implemented retry logic and a human-review queue for low-confidence extractions.
- Containerized the service with Docker and exposed it via FastAPI for internal batch processing.

Facial Sentiment Recognition | *Deep Learning, TensorFlow, Keras, CNN, OpenCV* GitHub August 2024

- Trained facial emotion recognition models on FER2013 (35,887 images, 7 classes) using VGG-16 and ResNet-50 transfer learning; achieving 88% test accuracy on 3-class subset (happy/neutral/sad), outperforming a baseline CNN by 12 percentage points.
- Built preprocessing pipeline with MTCNN face detection, grayscale normalization, and augmentation (rotation, horizontal flip, brightness jitter) to improve robustness across lighting conditions.

Achievements & Publications

SUI-IEEE Student Branch Chair led 30+ members, organized 12+ technical events (2025-26)
Best Paper Award in ACROSET 2024 on AI-Based Smart Education System to Enhanced the Learning of Students
Published chapters in IEEE Xplore & IGI Global AI Marketing and Ethics (2023)
Best Paper Presentation in IEEE ICPS-ITI-AGS on Tackling the Complexities of Federated Learning (2023)
Published Patent in Autonomous Drone Transportation Technology (2026)